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An orientation-dependent creep life evaluation method for nickel-based single crystal superalloys

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Abstract This paper aims to propose a creep life evaluation method considering the effect of crystallographic orientation. First, the maximum Schmid factor of $\{111\}$ $\langle 112 \rangle$ and the corresponding lattice rotation angle were introduced to form an “orientation factor”. Then the equivalent stress was calculated by multiplying this factor and the nominal stress. Latterly, the Larson-Miller Parameter (LMP) method was adopted as the rupture life evaluation criteria, in which the input variable was the equivalent stress instead of the nominal stress. All the predictions showed high accuracy when the proposed method was applied to Mar-M247, SC7-14, CMSX-2, Alloy454, CMSX-4 and DD6. Finally, the applicable temperature ranges of the orientation-dependent method (using equivalent stress) and the traditional LMP method (using nominal stress) were discussed. The results show that only the Orientation-Dependent (OD) method is reliable at intermediate temperatures (760–850 °C) because the orientation has significant effect on the stress rupture life, while the influence of orientation is considerably reduced at high temperatures. Both methods provide precise predictions in this situation, and the LMP method should be favored since it is much easier to implement.

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1. Introduction

The turbine entry temperature of advanced aircraft engines has reached approximately 2000 K due to the continuous push for greater thrust and higher efficiency.^{1,2} To achieve this extreme temperature, Single Crystal (SC) superalloys were widely used

in turbine blades for their superior high-temperature mechanical properties.^{3,4} Numerous scholars have recognized the vital role of creep rupture in the failure process of actual blades.⁵ Therefore, one of the design criteria of turbine blade was to provide sufficient creep/stress rupture life.

The issue of creep life evaluation has received considerable attention since 1950s. Engineers have tended to find a mathematical expression to describe the stress rupture life as a function of temperature and stress. Among these approaches, Larson-Miller Parameter (LMP) method,⁶ Manson-Haferd (MH) method⁷ and Dorn-Sherby-Orr (DSO) method⁸ are the best-known ones. These methods were based on the time-temperature relationships, and they could predict the rupture life of superalloys accurately in a certain range. Monkman and

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Grant⁹ reported that the product of the minimum creep strain rate and the stress rupture life was a constant, and this relationship was quite simple and effective.¹⁰ Kachanov¹¹ and Rabotnov¹² developed Continuum Damage Mechanics (CDM) method to account for the tertiary creep damage behavior of materials, and the creep rupture time predictions provided by CDM approach showed improved accuracy.¹³ There were also other methods for assessing the creep lives of isotropic materials, such as the Omega method,¹⁴ Theta method,¹⁵ and Wilshire method.^{16,17}

The creep strength of SC superalloys has been found significantly sensitive to the crystallographic orientation.^{18,19} In the case of the same loading condition, the stress rupture lives may differ a lot, even hundreds of times, due to the orientation effect. However, the application of aforementioned traditional methods is restricted to isotropic materials, and the orientation effect on the rupture life of SC superalloys has not been considered yet. Although massive data about rupture properties with different orientations was accumulated, the rupture strength design curve could not be built by these orientation-independent methods. The experimental data obtained at an extremely high cost could not directly contribute to engineering design. Up to now, the investigations on the effect of orientation on the stress rupture properties have been limited.

Sass and Feller-Kniepmeier²⁰ investigated the creep deformation of CMSX-4, and they reported that the minimum-creep-rate of orientations near $\langle 111 \rangle$ was the highest and the influence of orientation was obviously reduced as the temperature increased. Caron et al.²¹ analyzed the anisotropic creep response of several SC superalloys (CMSX-2, Alloy454, CMSX-4). Their results showed that the creep behavior was extremely sensitive to the orientation and the precipitate size at intermediate temperatures. More recently, experiments^{19,22–25} have established that single crystals with an asymmetric loading direction would undergo a time-dependent lattice rotation process, which could have a significant impact on the rupture strength. MacKay and Maier²⁶ investigated the influence of orientation on the stress rupture properties of MAR-M247. They claimed that the creep anisotropy of the SC superalloys could be attributed to dislocation slip and lattice rotation. Although there were some qualitative experimental researches about the effect of orientation on creep strength, the orientation-dependent rupture life evaluation methods have not been established yet.

The purpose of this investigation is to explore the probable mathematical expression among stress, rupture life and orientation at a certain temperature. Inspired by the function between rupture properties and orientation proposed by MacKay and Maier,²⁶ the Schmid factor and the lattice rotation angle are introduced to modify the nominal stress. Finally, the Orientation-Dependent (OD) method is validated using numerous experimental data of several SC superalloys with different orientations in Refs. 21,26,30.

2. Method

2.1. Calculation of crystallographic orientation

Experimental data of several SC superalloys with different orientations was collected from Refs. 21,26,30. As shown in Fig. 1(a), z_1 – z_7 represent the orientations of different specimens, and

these orientations are presented in the standard stereographic triangle. Each point in Fig. 1(a) represents an orientation of a certain specimen. To figure out the orientation of the specimens, the coordinates of these points must be calibrated at first.

Points “ j ”, “ k ” and “ l ” in Fig. 1(a) represent orientations $[001]$, $[011]$, $[11\bar{1}]$ respectively. If the length of the circle’s radius in Fig. 1(b) was assumed to be unit 1, polar coordinates of points “ j ”, “ k ” and “ l ” could be given as

$$j(0,0), k(\sqrt{2}-1, \frac{\pi}{2}), l(\frac{\sqrt{6}-\sqrt{2}}{2}, \frac{3}{4}\pi) \quad (1)$$

Based on Eq. (1), the position of other points in standard stereographic triangle was carefully measured.

Subsequently, the polar coordinate of these points was transferred to the orientation according to the principle of stereographic projection. Sample $z_1(r_z, \varphi_z)$ in Fig. 1(a) was taken as an example to illustrate this process. As shown in Fig. 1(c), r_z and φ_z indicate the length of line segment $||oz||$ and the magnitude of angle $\angle Xoz$. The Cartesian coordinates of point $b(u, v, w)$ in Fig. 1(c) could be calculated based on the triangle similarity relationship between $\triangle ozc$ and $\triangle abc$ in Fig. 1(c):

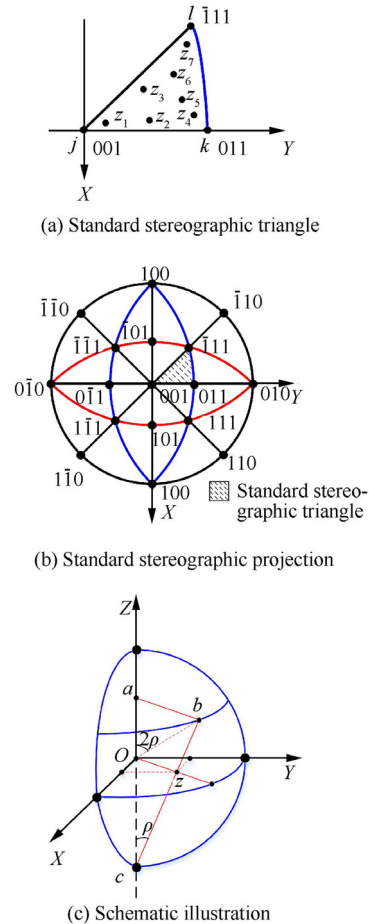


Fig. 1 Cubic crystal [001] standard stereographic projection.

$$\begin{cases} \rho = \arctan r_z \\ u = \sin(2\rho)\cos\varphi_z \\ v = \sin(2\rho)\sin\varphi_z \\ w = \cos(2\rho) \end{cases} \quad (2)$$

$[u, v, w]$ represents the crystallographic orientation of sample z_1 , and the orientations of the other specimens were determined using the same method.

2.2. Formation of equivalent stress in OD method

2.2.1. Effect of dislocation glide on $\{111\} \langle 112 \rangle$

The maximum Schmid factor of $\{111\} \langle 112 \rangle$ was selected as one of the control variables to form the “orientation factor”, because its magnitude could roughly reflect the creep anisotropy according to the experiments.²⁶ And dislocation motion on $\{111\} \langle 112 \rangle$ is also an important source of the inelastic deformation.

Dislocation glide is one of the major deformation mechanisms for SC superalloys. The Schmid factor reflects the relative magnitude of the resolved shear stress among different slip systems. $\{111\} \langle 110 \rangle$, $\{001\} \langle 110 \rangle$ and $\{111\} \langle 112 \rangle$ are the three slip systems which are commonly considered during the deformation of SC superalloys. Fig. 2(a), (b) and (c) show the maximum Schmid factor of these slip systems. Fig. 2(d) indicates the creep performance as a function of orientation at about 760 °C.²⁶ As we can see, the maximum Schmid factor of $\{111\} \langle 112 \rangle$ can roughly reflect the anisotropy of the stress rupture performance.

The dislocation slip on $\{111\} \langle 112 \rangle$ also serves as an important creep mechanism. Nickel-based SC superalloys mainly consist of two phases. The gamma (γ) phase is the matrix phase that presents a Face-Centered Cubic (FCC) structure. The gamma prime (γ') phase, which accounts for about 70% of

the volume, is the strengthening phase showing an ordered $L1_2$ structure. Although dislocation motion in these two phases is different, $\{111\} \langle 112 \rangle$ slip systems are important for both the γ phase and the γ' phase.

In a macroscopic sense, the slip system in an FCC metal is $\{111\} \langle 110 \rangle$. However, a dislocation on a $\{111\}$ plane of gamma phase dissociates into two Shockley partial dislocations, and the reaction is of the form

$$\frac{a_0}{2} [110](\bar{1}11) \rightarrow \frac{a_0}{6} [211](\bar{1}11) + \frac{a_0}{6} [12\bar{1}](\bar{1}11) \quad (3)$$

where a_0 is the lattice constant of FCC crystals and the dislocations on the right side belong to $\{111\} \langle 112 \rangle$ slip system.³¹ Therefore, the glide of a $\{111\} \langle 110 \rangle$ in the FCC metal is achieved by the passage of two partial dislocations of $\{111\} \langle 112 \rangle$, as shown in Fig. 3(a).

If the applied stress is sufficiently high at low or intermediate temperatures, the dislocations of $\{111\} \langle 110 \rangle$ will interact at phase boundary and form the “dislocation ribbon”. Then the “ribbon” enters the gamma prime phase. A typical reaction at the interface region might be

$$\frac{a_0}{2} [011] + \frac{a_0}{2} [\bar{1}01] \rightarrow \frac{a_0}{3} [\bar{1}12] + \frac{a_0}{6} [\bar{1}12] \quad (4)$$

as we can see from Fig. 3(b), the “ribbon” shears the γ' phase in the form of $\{111\} \langle 112 \rangle$ dislocations. The details of this process have been described by Reed et al.^{31,32}

Consequently, the dislocation motion of $\{111\} \langle 112 \rangle$ is an important creep mechanism for both γ and γ' phase, so it is reasonable to take the maximum Schmid factor of $\{111\} \langle 112 \rangle$ as the control variable in this method. However, there was still some phenomena that could not be explained by the Schmid factor. For example, although $[001]$ and $[011]$ have the same resolved shear stress, the creep performance of $[011]$ orientation is not as good as $[001]$. In addition, single crystals oriented

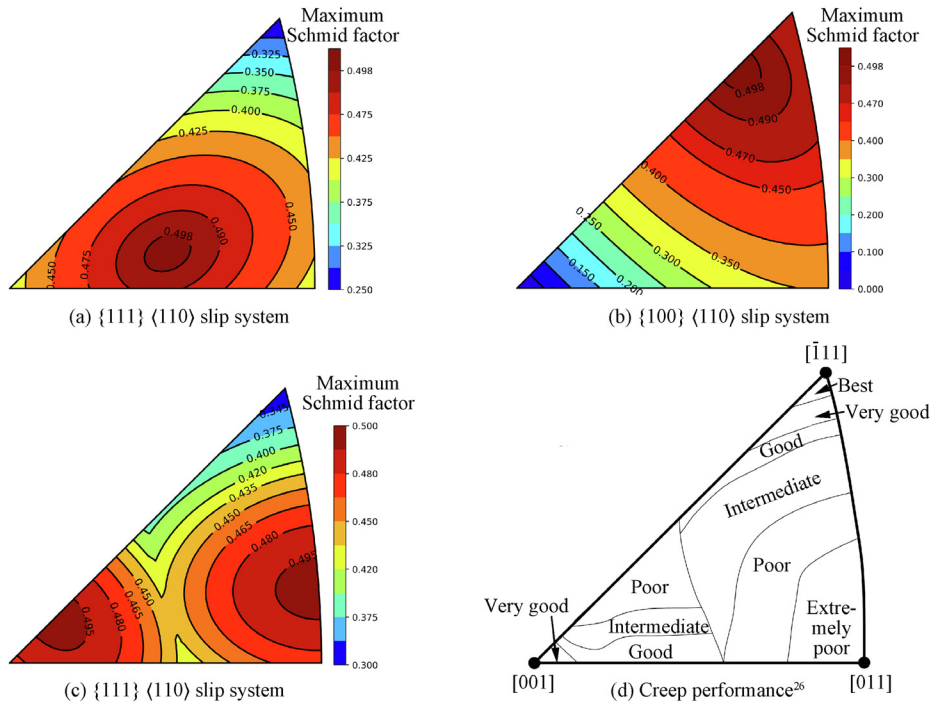
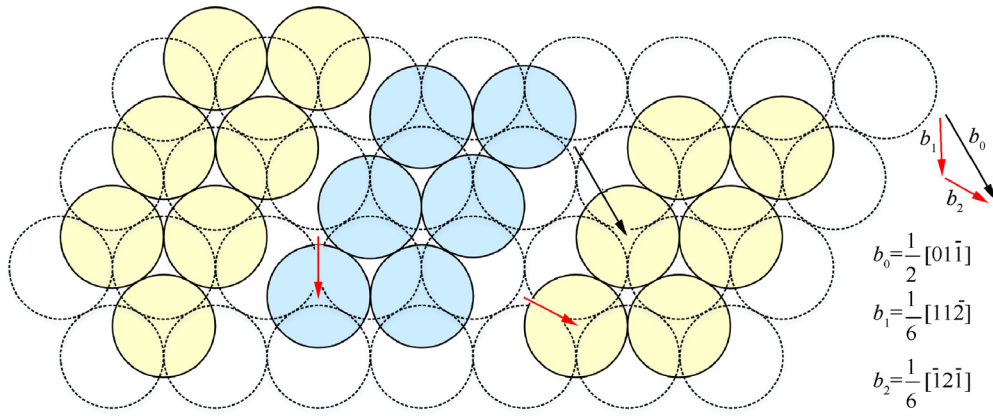
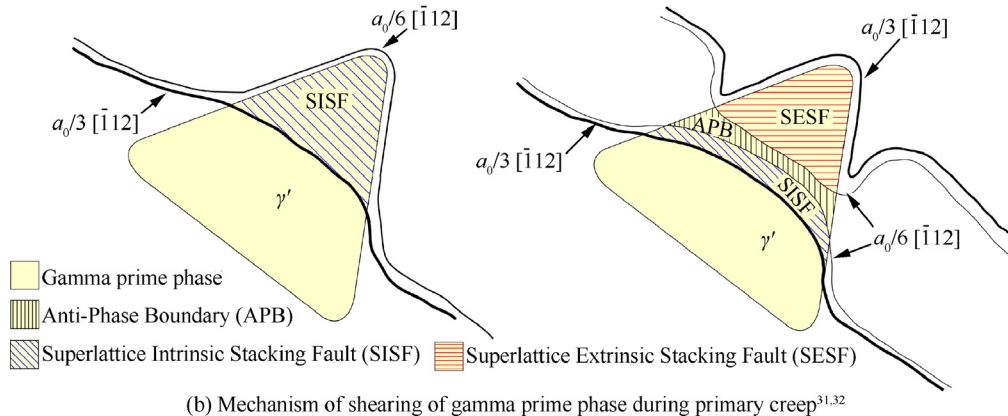


Fig. 2 Maximum Schmid factor of different slip systems and creep performance as a function of orientation at about 760 °C.²⁶



(a) A dislocation in gamma phase decomposed into two Shockley partial dislocations³¹



(b) Mechanism of shearing of gamma prime phase during primary creep^{31,32}

Fig. 3 Primary creep deformation mechanism of nickel-based single crystal superalloys.

near [001] exhibit significant differences in the rupture properties with similar Schmid factors according to Fig. 2(c) and 2(d). Therefore, some other control variables and parameters should be introduced in the method.

2.2.2. Lattice rotation process

During the process of creep deformation, the lattice of a single crystal specimen will rotate relative to a fixed coordinate frame.^{33,34} Extensive researches^{23–26} confirmed that the lattice rotation process had a significant effect on the stress rupture lives. Therefore, the lattice rotation angle was defined as the other control variable in the OD method.

Fig. 4(a) shows an SC sample under the tensile loading, σ is nominal stress, n is the normal direction of slip plane, s is the slip direction, the angle between the axis normal to slip plane and the loading axis is denoted by λ , and the angle between the slip direction and the loading axis is denoted by ϕ . The loading axis in the laboratory frame was defined as the “tensile axis”. It holds fixed because of the constraint of experimental equipment. The loading axis in the crystal frame was defined as the “tensile orientation”. It kept changing due to the lattice rotation. As the inelastic deformation occurs, the slip direction constantly turns to the “tensile axis”, and it is suggested that the “tensile orientation” gradually changes to the slip direction. When it comes to the single crystal samples under the

compressive loading, the normal direction of the slip plane would rotate toward the compressive axis,³³ as shown in Fig. 4(b).

The tensile orientation of a sample could be represented as a point in the stereographic projection triangle. Therefore, the change of tensile orientation could be traced by a series of points in the standard triangle. According to Schmid law, the slip system with maximum resolved shear stress was active. If $\{111\} \langle 112 \rangle$ slip systems were considered, there would be two distinct regions in the standard stereographic triangle, as shown in Fig. 5(a). $[211](111)$ is the only active slip system within area “ $lmnk$ ”, and $[112](\bar{1}11)$ is the active one in area “ jmn ”. These areas where only one slip system is active are called the “single slip regions”. The “ lm ” and “ jn ” boundaries represent the duplex slip regions, and the Schmid factors are equivalent for at least two different slip systems in these regions. Points “ j ” and “ l ” indicate the high symmetrical orientations. More than three slip systems would be equally stressed in the two points, meaning that dislocation slip could occur simultaneously on different slip systems.

MacKay and Maier²⁶ summarized the lattice evolution process of Mar-M247 in 1982. Many other experimental observations^{19,22,24,25} at intermediate temperatures supported their research. The creep sample oriented in “ g ” was taken as an

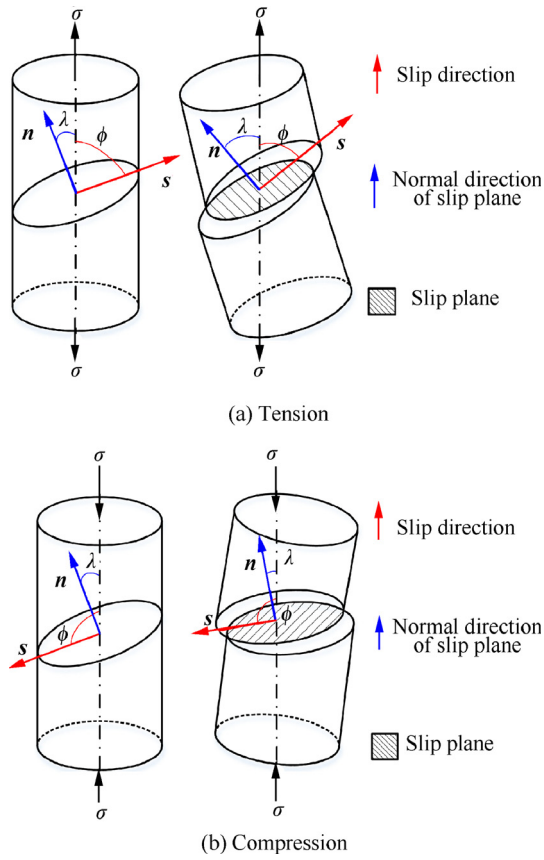


Fig. 4 Illustration of single crystal material's deformation under uniaxial loading.³³

example to illustrate this process. As shown by the arrows in Fig. 5(a), the lattice of the sample “g” would rotate until the duplex slip boundaries (“lm”) were reached or nearly reached (“g”→“h”). Then the tensile orientation gradually changed along the “lm” boundary (“h”→“l”). Finally, the orientation stopped changing near point “l”, since the lattice was highly symmetrical there. The sample might fail before reaching point “l” or the “lm” boundary. A similar process also occurred for SC superalloys oriented within the “jmn” region. For the purpose of simplicity, the path of the changing tensile orientation was approximated to a straight line in the standard triangle. This assumption was roughly consistent with some experiments.^{19,22,24–26}

MacKay and Maier²⁶ found that the SC superalloys exhibit shorter rupture lives if they rotate greater angles to reach the duplex slip boundaries. In other words, the effect of lattice rotation on creep strength in the “jn” or “lm” boundaries is not as significant as in the single slip region, because the interaction among slip systems leads to sufficient strain hardening in the duplex slip region. As a result, the lattice rotation angle to reach “jn” or “lm” boundary was chosen as the other control variable in the OD method. This variable was just the angle between the initial tensile orientation and the corresponding tensile orientation on the boundaries. For instance, the angle between orientation $g([u_g, v_g, w_g])$ and orientation $h([u_h, v_h, w_h])$ in the crystal frame is the lattice rotation angle for sample “g”. Moreover, the rotation angle for other points in the standard triangle is shown in Fig. 5(b).

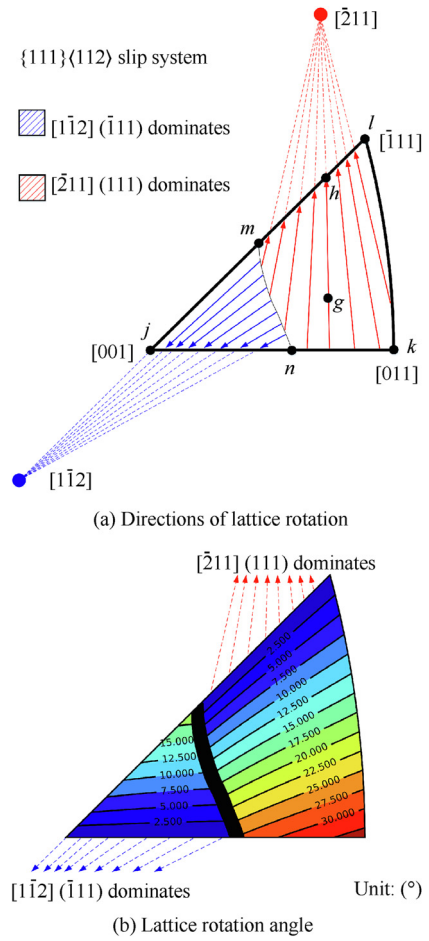


Fig. 5 Illustration of lattice rotation process in standard triangle.

2.2.3. Orientation factor

In order to account for the effect of orientation, an “orientation factor” was introduced to modify the nominal stress in the OD method. The maximum Schmid factor of $\{111\} \langle 112 \rangle$ and the lattice rotation angle were utilized as the control variables, and another two parameters were used to form the “orientation factor” as

$$\begin{cases} f = (M^{\max})^{k_1} + k_2 \theta \\ \bar{\sigma} = \sigma f \end{cases} \quad (5)$$

where f is the “orientation factor”; $\bar{\sigma}$ is the equivalent stress; $M^{\max}$ is the maximum Schmid factor of $\{111\} \langle 112 \rangle$; k_1 is a temperature-dependent parameter, and $(M^{\max})^{k_1}$ reflects the influence of dislocations glide, as the absolute value of k_1 approaches 0, the effect of dislocation glide on rupture life becomes insignificant; θ is the lattice rotation angle (radian) to reach the duplex slip boundaries; k_2 is also a temperature-dependent parameter, which characterizes the effect of lattice rotation, as the absolute value of k_2 approaches 0, the effect of lattice rotation is negligible similarly.

If k_1 and k_2 are both 0, the “orientation factor” will be unit 1. Then the equivalent stress is just the nominal stress, and the OD method becomes the traditional LMP method.

Table 1 Experimental data summary of SC superalloys at intermediate temperature.

Superalloy	Experimental condition	Orientation number	Source
Mar-M247	774 °C, 724 MPa	22 (shown in Fig. 6)	Ref. 26
SC7-14	760 °C, 689/758 MPa	4 (shown in Fig. 7)	Ref. 27
CMSX-2	760 °C, 750 MPa (γ' size is 0.30 μm)	13 (shown in Fig. 8)	Ref. 21

2.3. Larson-Miller parameter method

The traditional Larson-Miller parameter method takes the nominal stress as the control variable to assess rupture lives of isotropic materials. The commonly used equation is shown as

$$b_0 + b_1 \lg \sigma + b_2 \lg^2 \sigma + \dots + b_n \lg^n \sigma = T(C + \lg t_r) \quad (6)$$

where T is the temperature in Kelvin; t_r is creep rupture life; b_i and C are material parameters obtained by optimization.

The relationship of third-degree polynomial between $\lg \sigma$ and $\lg t_r$ was adopted, and the change of temperature was not considered here for simplicity. If the equivalent stress was used in the LMP equation instead of the nominal stress, the stress rupture lives of SC superalloys with different tensile orientations could be assessed through

$$\begin{cases} \lg t_r = d_1 + d_2 x + d_3 x^2 + d_4 x^3 \\ x = \lg \bar{\sigma} \end{cases} \quad (7)$$

where d_i are the temperature-dependent parameters determined by fitness of experimental data.

3. Results and discussion

3.1. Application at intermediate temperatures (760–850 °C)

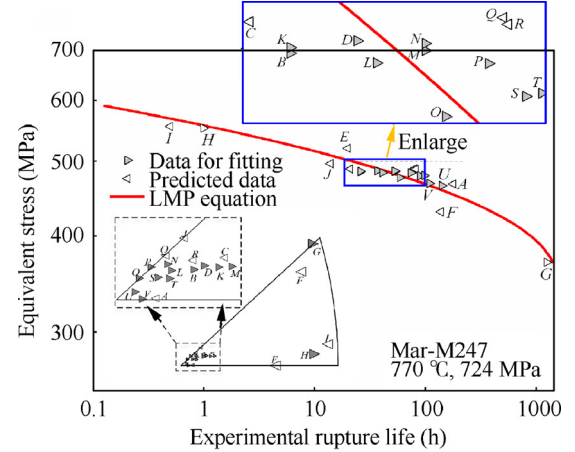
The influence of crystallographic orientation is very significant and the lattice rotation process observed^{19,22,24,26} was roughly consistent with Fig. 5(a) at intermediate temperatures. In order to verify the applicability of the OD method at these temperatures, numerous experimental data about rupture lives with different orientations was collected. The data was listed in Table 1.

Part of the experimental data (shown in Figs. 6 and 8) was used to fit the coefficients in Eqs. (5) and (7) with the help of MATLAB LSQCURVEFIT toolbox. The experimental data chosen to fit the parameters roughly covered the entire range of rupture lives and the orientations. The error function was defined as

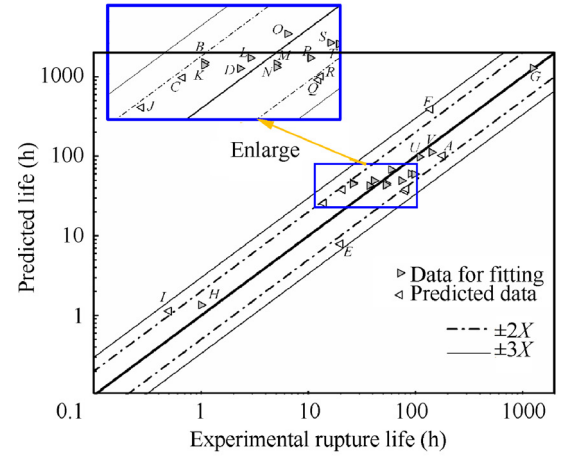
$$\text{error} = \sum_{i=1}^{i=n_i} (|t_i^{\text{res}} - t_i^{\text{exp}}| / t_i^{\text{exp}}) W_i \quad (8)$$

where t_i^{exp} is the experimental stress rupture time; t_i^{res} is the life estimated by the OD method; n_i is the number of the samples; W_i is a weighting parameter of the i th data.

An optimization function was used to minimize errors. The obtained material parameters are listed in Table 2, and Figs. 6–8 show the results at intermediate temperatures. The data chosen to fit the coefficients is denoted as the “data for fitting” in Figs. 6 and 8, then the creep lives of the other samples were predicted based on these coefficients, and these predicted lives are denoted as the “predicted data” in Figs. 6 and 8.



(a) Polynomial regression curve



(b) Predicted creep rupture lives

Fig. 6 Creep rupture prediction of Mar-M247²⁶ at 774 °C, 724 MPa.

There are 22 samples (“A”–“V”) under 774 °C, 724 MPa in Fig. 6. All the samples had different tensile orientations. The stress rupture life of sample “I” and sample “G” differed by thousands of times under the same loading condition, and it was suggested that crystal orientation had a significant effect on the creep strength of Mar-M247 at 774 °C. If the nominal stress was chosen as the input variable of the LMP method, all the predicted rupture lives would be the same because of the same loading condition. Apparently, the traditional LMP method could not be used to evaluate the rupture lives at intermediate temperatures. When the equivalent stress was used, an “orientation factor” would be imposed to account for the effect of orientation. As shown in Fig. 6(a), the difference of the equivalent stress among these samples exceeds 200 MPa.

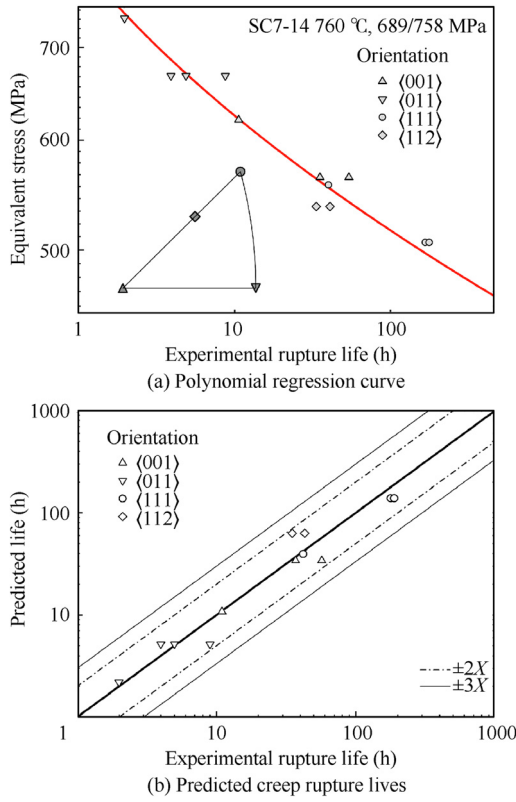


Fig. 7 Creep rupture prediction of SC7-14²⁷ at 760 °C.

The influence of crystal orientation was reflected by the difference of equivalent stress. Therefore, the OD method could consider the effect of orientation and be used to evaluate creep lives of SC superalloys at intermediate temperatures.

Fig. 6(b) shows the creep rupture predictions of Mar-M247. It was observed that the rupture time predictions under 774 °C, 724 MPa were within a factor of three, and the experimental data used to fit material parameters was in a factor of 2. The OD method showed high accuracy when it was applied to Mar-M247.

Fig. 7 indicates the results of SC7-14 at 760 °C. There are 12 samples with 4 different orientations. The creep behavior was highly anisotropic. The $\langle 111 \rangle$ orientation exhibited the best creep strength. The rupture lives of the specimens oriented near $\langle 111 \rangle$ were about thirty times longer than those oriented near $\langle 011 \rangle$. The relationship between creep performance and crystal orientation agreed well with the function proposed by MacKay and Maier.²⁶ Compared to the experimental lives, all the predictions were within a factor of 2.

The method also worked well in the application to CMSX-2. Caron et al.²¹ prepared 13 samples with γ' size of 0.30 μm . Every sample had its unique orientation, and they were tested under the same creep loading condition (760 °C, 750 MPa). However, the rupture lives of different specimens could differ by hundreds of times. The experimental results in Fig. 8 further confirmed that orientation had a significant influence on creep strength at intermediate temperatures, and the LMP method was not suitable for the materials whose rupture lives were extremely sensitive to orientation.

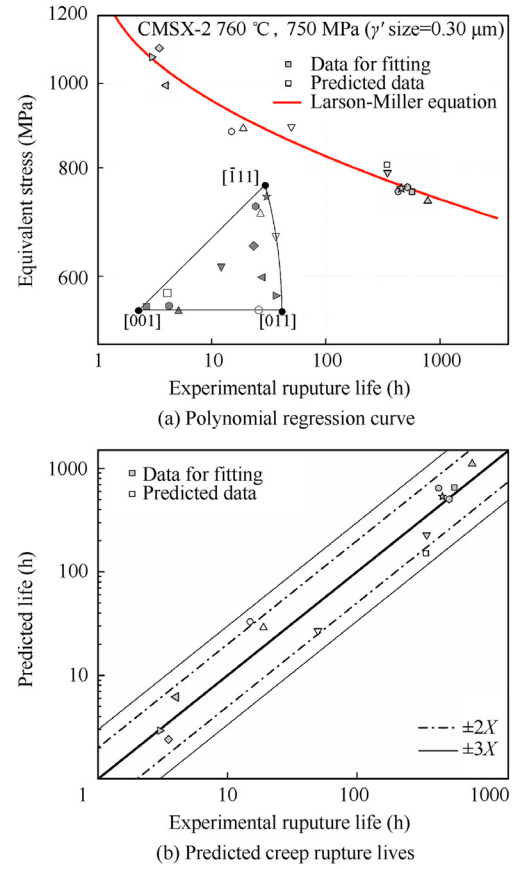


Fig. 8 Creep rupture prediction of CMSX-2²¹ at 760 °C, 750 MPa.

3.2. Application at high temperatures

The creep anisotropy was significantly reduced at high temperatures and the lattice rotation process was not aligned exactly with Fig. 5(a).^{23,29} It could be attributed to the deformation mechanisms that activated at higher temperatures. To verify the applicability of the OD method, a large amount of experimental data was collected. The information about these creep samples is shown in Table 3.

To quantitatively measure the accuracy of the method, the coefficient of determination (R^2) at each temperature was calculated as

$$R^2 = 1 - \frac{\sum_i (t_i^{\text{exp}} - \bar{t}^{\text{exp}})^2}{\sum_i (t_i^{\text{exp}} - \bar{t}^{\text{exp}})^2} \quad (9)$$

where $\bar{t}^{\text{exp}}$ is the average experimental rupture time of all the samples. $R^2 \in [0, 1]$, the closer R^2 is to 1, the more accurate the method is. When $R^2 = 1$, all the experimental points will fall just on the design curve.

MacLachlan et al.²⁹ investigated the anisotropic creep behavior of CMSX-4 at 950 °C, and the rupture time predictions using the LMP method are presented in Fig. 9(a). It was apparent that orientation had little effect on the rupture properties of CMSX-4 at 950 °C. The scatter factor of the samples with the same loading condition did not exceed twice. This result may be explained by the fact that more deformation

Table 2 Coefficients of OD method for Mar-M247, SC7-14-6, and CMSX-2 at intermediate temperatures.

Material	Temperature (°C)	d_1	d_2	d_3	d_4	k_1	k_2
Mar-M247	774	-247.37	93.66	42.20	-15.84	0.604	0.237
SC7-14-6	760	84.39	-22.549	-11.979	3.359	0.267	0.255
CMSX-2	760	150.75	1.20	-47.05	10.00	0.0327	0.882

Table 3 Experimental data summary of SC superalloys at high temperatures.

Superalloy	Experimental condition	Orientation	Source
CMSX-4	950 °C, 185–450 MPa	6 (shown in Fig. 9(a))	Ref. 29
SC7-14	980 °C, 248 MPa	4 (shown in Fig. 10)	Ref. 27
CMSX-2	1050 °C, 120 MPa (γ' size is 0.30 μm)	8 (shown in Fig. 10)	Ref. 21
CMSX-2	1050 °C, 120 MPa (γ' size is 0.45 μm)	10 (shown in Fig. 10)	Ref. 21
Alloy 454	980 °C, 200 MPa (γ' size is 0.50 μm)	9 (shown in Fig. 10)	Ref. 21

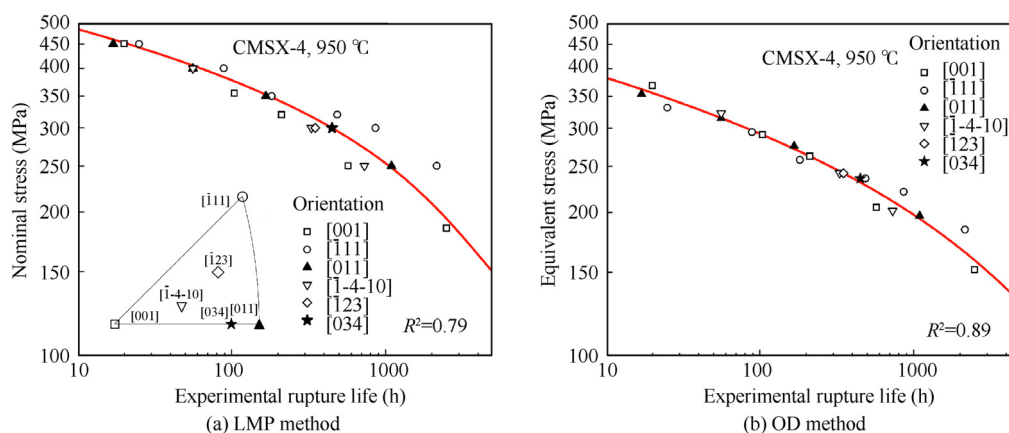
mechanisms were activated at elevated temperatures.^{26,29} Sass and Feller-Kniepmeier,²⁰ and Caron et al.²¹ also believed that γ' rafting at high temperatures may have some contribution to this phenomenon. Therefore, with the increase of temperature, the effect of crystallographic orientation on the creep strength was gradually weakened. The rupture properties of SC superalloys turned to be isotropic. R^2 of the polynomial regression curve using the nominal stress was 0.79. When the equivalent stress was adopted, R^2 was changed to 0.89. No significant improvement was found as shown in Fig. 9(b). Consequently, both LMP method and OD method show high accuracy in the application to CMSX-4 at 950 °C.

Furthermore, Fig. 10 illustrates the experimental results of SC7-14, Alloy454, CMSX-2 at high temperatures. These SC superalloys were tested under the condition of 980 °C/248 MPa, 980 °C/200 MPa, 1050 °C/120 MPa respectively, and their orientations were presented in the standard stereographic triangle. The LMP method was insufficient to evaluate the creep lives of SC7-14, Alloy454 and CMSX-2 in this situation, because the nominal stress made no difference. Therefore, the stress rupture lives of the samples were numerically averaged, and the averaged rupture time was regarded as the predicted lives. In addition, the predicted rupture time of CMSX-4 by LMP method is also shown in Fig. 10.

There were 23 samples with 4 different orientations for SC7-14 at 980 °C. $\langle 111 \rangle$ showed the best creep performance. The difference of rupture lives among specimen oriented near $\langle 111 \rangle$ and $\langle 011 \rangle$ was less than ten times. Orientation still had some impact on rupture lives, but not as significant as at 760 °C. The results of Alloy454 contained 9 samples with different orientations. All the specimens were under the test condition of 980 °C, 200 MPa. The interesting aspect was that they had almost the same rupture lives. The ratio of the longest life to the shortest one was less than twice. The orientation hardly affected the creep strength of Alloy454 at 980 °C. All the rupture predictions of CMSX-4, SC7-14, Alloy454 and CMSX-2 were within a factor of 3. They were all estimated using the orientation-independent methods. In other words, the LMP method was effective in the application to these superalloys at high temperatures.

Deformation mechanisms such as cross-slip and climbing were activated at high temperature. However, the creep deformation mechanism at intermediate temperatures was mainly the dislocation slip on the $\{111\}$ $\langle 112 \rangle$ slip systems. Therefore, the creep performance of single crystal superalloys gradually turned to be isotropic as the temperature increased.

The OD method was also used to assess the stress rupture lives of these single crystals at high temperatures, as shown

**Fig. 9** Creep rupture predictions of CMSX-4²⁹ at 950 °C.

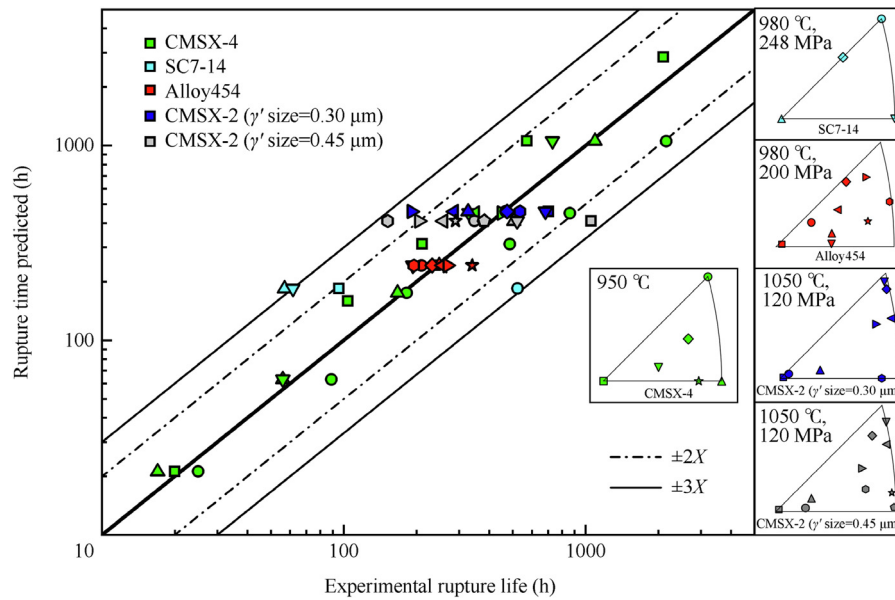


Fig. 10 Creep rupture predictions of CMSX-4,²⁹ SC7-14,²⁷ Alloy454,²¹ CMSX-2²¹ using orientation-independent method at high temperatures.

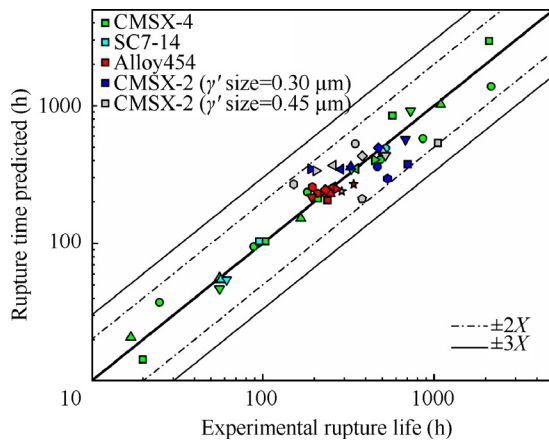


Fig. 11 Creep rupture predictions of CMSX-4,²⁹ SC7-14,²⁷ Alloy454,²¹ CMSX-2²¹ using OD method at high temperatures.

in Fig. 11. The coefficient fitting strategy was aforementioned in Section 3.1, and the parameters of the OD method at high temperatures are summarized in Table 4. It should be noted that part of k_1 and k_2 in Table 4 was negative, because the basis of the OD method did not match the creep mechanism

at high temperatures. The inspiration of OD method came from the function between rupture properties and orientation proposed by MacKay and Maier.²⁶ However, this function was not suitable for all the SC superalloys, and the influence of temperature on the creep anisotropy was also not considered. The absolute value of the parameter k_1 and the parameter k_2 in Table 4 was quite close to 0, which meant that the “orientation factor” was close to 1 at high temperatures. The equivalent stress and the nominal stress were almost the same in this situation. In other words, the OD method was quite similar to the LMP method in this situation.

Fig. 11 illustrates the predicted lives versus the experimental results in detail. It was observed that all the predicted results fall into the error band of 2. Although the proposed OD method could match the existing data better compared to the LMP method, it took more efforts to implement and its basis was not totally consistent with the creep mechanism at high temperatures. Therefore, it was more favored to use the LMP method at high temperatures.

3.3. Applicable temperature ranges of OD method

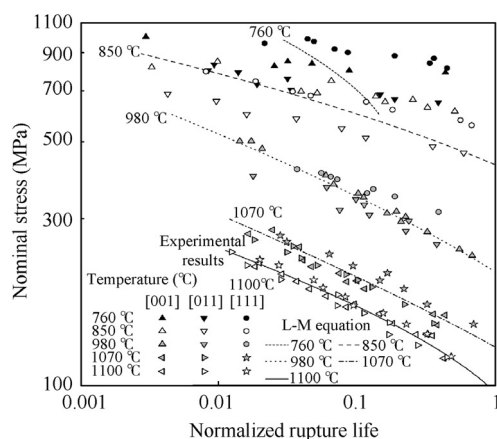
The effect of orientation on stress rupture properties was temperature-dependent. It was not notable at high temperatures, and both the LMP and OD method could exhibit some

Table 4 Coefficients of OD method for SC7-14, CMSX-2 and Alloy 454 at high temperatures.

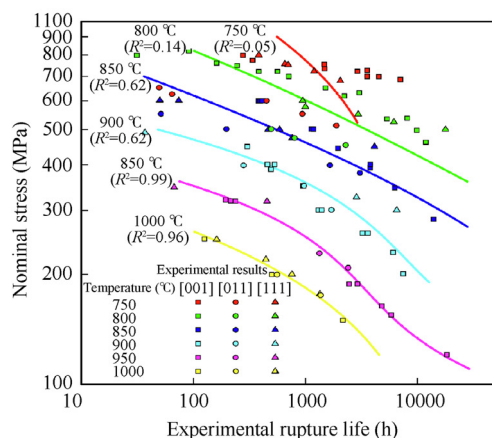
Material	Temperature (°C)	d_1	d_2	d_3	d_4	k_1	k_2
CMSX-4	950	62.98	-80.96	40.17	-6.71	0.2645	-0.0548
SC7-14	980	99.84	-103.21	31.69	-2.28	1.0400	0
CMSX-2	1050 (γ' size = 0.30 μm)	86.67	-28.55	-24.60	9.03	0.0840	0.0329
CMSX-2	1050 (γ' size = 0.45 μm)	217.83	-64.20	-76.87	27.90	-0.0134	0.0999
Alloy 454	980	19.40	10.37	1.96	-4.24	0.0190	-0.0159

accuracy. While at intermediate temperatures, the rupture time of samples with different orientation could differ by hundreds of times, as shown in Figs. 6–8. Only the OD method was applicable in this situation. However, the OD method was not as simple as the LMP method. It required the calculation of the Schmid factor and the lattice rotation angle. Therefore, engineers needed to strike a balance between the accuracy and the simplicity. In order to figure out the suitable temperature range of the OD method, the experimental data about DD6²⁸ and CMSX-4³⁰ was collected from literatures.

The rupture time predictions of DD6 provided by the traditional LMP method are shown in Fig. 12(a). All the experimental data with different orientations was used to fit the coefficients. Apparently, the rupture strength was orientation-dependent at 760 °C and 850 °C. It was easy to distinguish the experimental points of different orientations at these temperatures. [111] exhibited significantly better creep properties at 760 °C, while [001] showed the best performance at 850 °C. Closer inspection of Fig. 12(a) showed that stress rupture properties were still slightly influenced by orientation at 980 °C. While at 1070 °C and 1100 °C, there was almost no difference in the rupture properties among these three orientations. As a result, the LMP method showed high accuracy at 1070 °C and 1100 °C. And it also worked well at 980 °C. However, the rupture predictions were not acceptable at 760 °C and 850 °C.



(a) DD6²⁸



(b) CMSX-4³⁰

Fig. 12 Creep rupture evaluation using LMP method at different temperatures.

Moreover, the rupture life predictions of CMSX-4 provided by LMP method are presented in Fig. 12(b). The results of CMSX-4 were similar to DD6. The rupture time was extremely sensitive to orientation at temperatures below 950 °C. The creep life of the samples with different orientations could differ by hundreds of time, although they were under the same loading condition. [011] exhibited the worst rupture properties among the typical orientations at 750 °C, 800 °C and 850 °C, while the performance of the samples with different orientations was not diverse significantly at 950 °C and 1000 °C. As the temperature rose from 750 °C to 950 °C, there had been a gradual increase of R^2 as shown in Fig. 12(b). R^2 was close to 1 at the temperatures above 950 °C, meaning that the rupture time could be predicted precisely by the LMP method in these situations. However, at the temperatures from 750 °C to 900 °C, the error was not acceptable.

The OD method was used to make rupture time predictions about DD6, part of the coefficients can be found in Table 5, and the coefficient of determination of the LMP method and the OD method is indicated in Fig. 13. What was interesting in Fig. 13 was the continual growth of R^2 of the LMP method, it changed from 0.069 at 760 °C to 0.97 at 1100 °C, while R^2 of the OD method was always larger than 0.85. In other words, the predictions provided by OD method were much more precise than those provided by LMP method at 760 °C and 850 °C. However, there was no significant difference in accuracy between these two methods at temperatures above 980 °C.

Therefore, only the OD method was available at intermediate temperatures, because it considered the significant influence of the crystallographic orientation. However, the LMP method was more favorable at high temperatures, since it was easier to implement and could produce accurate predictions. The results of the LMP method were satisfactory in the application to DD6 and CMSX-4 at the temperatures above 950 °C.

Table 5 Coefficients of OD method for DD6 at different temperatures.

Temperature (°C)	760	850	980	1070	1100
k_1	0.307	-0.113	0.127	0.0976	0.0831
k_2	0.225	0.511	0.154	0.0466	0.0421

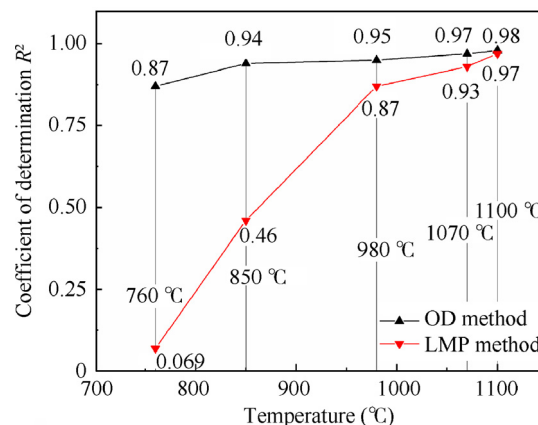


Fig. 13 Coefficient of determination of LMP and OD method for DD6 at different temperatures.

The OD method is applicable when dislocation slip on $\{111\}$ $\langle 112 \rangle$ was the dominant deformation mechanism, and the applicable temperature ranges should be determined by experiments for specific SC superalloys. Reed et al.^{31,32} have proved that dislocation slip on $\{111\}$ $\langle 112 \rangle$ was an important deformation mechanism of “the primary creep”, and the single crystal superalloys usually underwent significant primary creep deformation under the condition of “low/intermediate temperatures and high applied stress”. Consequently, it was believed that the OD method could also be used to evaluate the stress rupture lives of SC superalloys at low temperatures (400–700 °C). However, the most concerning temperature range about the creep of SC blade was above 700 °C, and the investigations about the creep strength at low temperatures were really limited. More experimental researches were needed to verify the application of this method at the temperature ranging from 400 °C to 700 °C.

4. Conclusions

In this paper, the Schmid factor and the lattice rotation angle were introduced to modify the nominal stress. An OD method was proposed to evaluate the stress rupture lives of SC superalloys. The following conclusions can be drawn:

- (1) The OD method has been successfully applied in the creep life evaluation of several SC superalloys including Mar-M247, CMSX-2, SC7-14, Alloy454, CMSX-4 and DD6. The effect of crystallographic orientation on the stress rupture lives at intermediate temperatures was successfully modeled by this OD method.
- (2) The influence of crystallographic orientation on rupture properties was extremely sensitive to temperature. The creep anisotropy was significantly reduced as the temperature increased. It has been quantitatively proved that the results of the traditional LMP method were pretty satisfactory in the application to CMSX-2, SC7-14-6, Alloy454, CMSX-4 and DD6 at high temperatures.
- (3) For DD6 and CMSX-4, the LMP method was more favorable at the temperatures above 950 °C, since it was easier to implement. And the OD method showed more advantages at intermediate temperatures (760–850 °C), because the influence of orientation could not be ignored. And the specific applicable temperature ranges of the OD method for other SC superalloys depended on their characteristics and should be determined by experiments.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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